

Slope, Parallel Lines and Perpendicular Lines

Answer Key

High School Geometry

Quick Answers

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| 1. $\frac{2}{3}$ | 8. (a) slope of AB = $1 \cdot \frac{1}{3}$, (b) slope of DC = $1 \cdot \frac{1}{3}$, |
| 2. $-\frac{3}{5}$ | (c) slope of AD = 3, — |
| 3. undefined | 9. (a) slope of PQ = $1 \cdot \frac{1}{3}$, (b) slope of QR = -3, |
| 4. (a) slope of j = 2, (b) slope of k = 2, — | — |
| 5. (a) slope of p = $1 \cdot \frac{1}{2}$, (b) slope of q = -2, — | 10. (a) slope of EF = $-\frac{1}{2}$, (b) slope of HG = $-\frac{1}{2}$, |
| 6. $-\frac{3}{4}$ | (c) slope of FG = 2, — |
| 7. $-\frac{1}{2}$ | |
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What is verified

5 of 10 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 4, 5, 8, 9, 10. The worked solution states what a correct response must contain.

Curriculum

Level: High School Geometry

HSG-GPE.B.4 – problems 8, 10

HSG-GPE.B.5 – problems 1, 2, 3, 4, 5, 6, 7, 9

Difficulty 1–5 of 5 across 22 tagged checks.

Common wrong answers

1. *If they got 1.5: wrote run over rise instead of rise over run*
 2. *If they got 0.6: subtracted the coordinates in opposite orders top and bottom, losing the minus sign*
 7. *If they got 0.5: took the reciprocal but forgot to negate it*
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- 1.** Find the slope of the line through (2, 3) and (8, 7).

Solution. Rise over run, both differences in the same order: $m = \frac{7-3}{8-2} = \frac{4}{6}$, which reduces,

so

$m = \frac{2}{3}$

- 2.** Find the slope of the line through (-4, 5) and (6, -1).

Solution. $m = \frac{-1 - 5}{6 - (-4)} = \frac{-6}{10}$, so

$$m = -\frac{3}{5}$$

3. Find the slope of the line through $(5, -2)$ and $(5, 9)$.

Solution. The run is $5 - 5 = 0$, so the slope would require division by zero: the line is vertical and has no slope.

$$\text{undefined}$$

4. Line j through $(-2, 1)$ and $(2, 9)$; line k through $(0, -5)$ and $(3, 1)$. (a) Slope of j . (b) Slope of k . (c) Parallel, perpendicular or neither?

Solution (a). $m_j = \frac{9 - 1}{2 - (-2)} = \frac{8}{4}$, so

$$m_j = 2$$

Solution (b). $m_k = \frac{1 - (-5)}{3 - 0} = \frac{6}{3}$, so

$$m_k = 2$$

Solution (c) — instructor-judged. A complete response says the slopes are equal, and that equal slopes mean the lines rise by the same amount for every unit of run, so they never meet: the lines are *parallel*. Naming them parallel without citing the equal slopes earns partial credit.

5. Line p through $(-1, 4)$ and $(3, 6)$; line q through $(2, -3)$ and $(4, -7)$. (a) Slope of p . (b) Slope of q . (c) Parallel, perpendicular or neither?

Solution (a). $m_p = \frac{6 - 4}{3 - (-1)} = \frac{2}{4}$, so

$$m_p = \frac{1}{2}$$

Solution (b). $m_q = \frac{-7 - (-3)}{4 - 2} = \frac{-4}{2}$, so

$$m_q = -2$$

Solution (c) — instructor-judged. A complete response multiplies the slopes, $\frac{1}{2} \times (-2) = -1$, and states that slopes whose product is -1 belong to perpendicular lines — equivalently that each slope is the negative reciprocal of the other. Full credit requires the product or the negative-reciprocal relationship to be shown, not just the word perpendicular.

6. Find the slope of any line parallel to the line through $(-6, 2)$ and $(2, -4)$.

Solution. Parallel lines share a slope, so compute the given line's: $m = \frac{-4 - 2}{2 - (-6)} = \frac{-6}{8}$, giving

$$m = -\frac{3}{4}$$

7. Find the slope of any line perpendicular to the line through $(3, -5)$ and $(9, 7)$.

Solution. The given line has slope $\frac{7 - (-5)}{9 - 3} = \frac{12}{6} = 2$. A perpendicular line has the negative reciprocal, which read straight off the coordinates is $-\frac{9 - 3}{7 - (-5)} = -\frac{6}{12}$, so

$$m = -\frac{1}{2}$$

8. $ABCD$ with $A(-1, 0)$, $B(5, 2)$, $C(7, 8)$, $D(1, 6)$. (a) Slope of $\overline{AB}$. (b) Slope of $\overline{DC}$. (c) Slope of $\overline{AD}$. (d) Why is $ABCD$ a parallelogram?

Solution (a). $\frac{2-0}{5-(-1)} = \frac{2}{6}$, so

$$m_{AB} = \frac{1}{3}$$

Solution (b). $\frac{8-6}{7-1} = \frac{2}{6}$, so

$$m_{DC} = \frac{1}{3}$$

Solution (c). $\frac{6-0}{1-(-1)} = \frac{6}{2}$, so

$$m_{AD} = 3$$

Solution (d) — instructor-judged. A complete response notes that $\overline{AB}$ and $\overline{DC}$ have equal slopes, computes $m_{BC} = \frac{8-2}{7-5} = 3$ and observes that it equals m_{AD} , so both pairs of opposite sides are parallel — which is the definition of a parallelogram. Full credit requires both pairs to be addressed.

9. Triangle PQR with $P(-2, 1)$, $Q(4, 3)$, $R(6, -3)$. (a) Slope of $\overline{PQ}$. (b) Slope of $\overline{QR}$. (c) Right-angled at Q ?

Solution (a). $\frac{3-1}{4-(-2)} = \frac{2}{6}$, so

$$m_{PQ} = \frac{1}{3}$$

Solution (b). $\frac{-3-3}{6-4} = \frac{-6}{2}$, so

$$m_{QR} = -3$$

Solution (c) — instructor-judged. A complete response multiplies the slopes of the two sides that meet at Q : $\frac{1}{3} \times (-3) = -1$, so $\overline{PQ} \perp \overline{QR}$ and the angle at Q is a right angle — the triangle is right-angled at Q . Full credit requires the product to be shown and tied to the vertex where the two sides meet.

10. $EFGH$ with $E(-3, -1)$, $F(1, -3)$, $G(5, 5)$, $H(1, 7)$. (a) Slope of $\overline{EF}$. (b) Slope of $\overline{HG}$. (c) Slope of $\overline{FG}$. (d) What kind of quadrilateral is $EFGH$?

Solution (a). $\frac{-3-(-1)}{1-(-3)} = \frac{-2}{4}$, so

$$m_{EF} = -\frac{1}{2}$$

Solution (b). $\frac{5-7}{5-1} = \frac{-2}{4}$, so

$$m_{HG} = -\frac{1}{2}$$

Solution (c). $\frac{5-(-3)}{5-1} = \frac{8}{4}$, so

$$m_{FG} = 2$$

Solution (d) — instructor-judged. A complete response uses the equal slopes of $\overline{EF}$ and $\overline{HG}$, computes $m_{EH} = \frac{7-(-1)}{1-(-3)} = 2$ and matches it to m_{FG} , so both pairs of opposite sides are parallel; then multiplies $m_{EF} \times m_{FG} = -\frac{1}{2} \times 2 = -1$, so adjacent sides meet at right angles. A parallelogram with right angles is a *rectangle*. Full credit requires both the parallel argument and the perpendicular argument.